

Akash Debbarma

+91 6009636806 | akashdebbarma.cse@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in machine learning, deep learning, and technical research. Proficient in Python, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SQL, and PostgreSQL. Experienced in building real-time applications with Next.js, React, TypeScript, Supabase, and WebSockets, with research exposure to blockchain scalability and Layer 2 technologies.

EDUCATION

Techno College of Engineering, Agartala Aug 2024 – Jun 2028
B.Tech in Computer Science and Engineering

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Statistics

TECHNICAL SKILLS

Programming: Python, C++, SQL, R
Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn
Data Science: Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Google Colab, Data Preprocessing, Feature Engineering, Statistical Analysis, Model Evaluation
AI: Machine Learning, Deep Learning, Neural Networks, NLP, Computer Vision, Regression, Classification
Web & Software: Next.js, React, TypeScript, Tailwind CSS, WebSockets, Full-Stack Development, Real-Time Systems
Databases & Tools: PostgreSQL, Supabase, Git, GitHub, VS Code

PROJECTS

Aether Chat – Real-Time Communication Platform Aug 2026 – Present
Next.js, React, TypeScript, Supabase, PostgreSQL, WebSockets GitHub

- Architected a full-stack real-time messaging platform supporting one-to-one and group conversations with WebSocket-based synchronization.
- Engineered a PostgreSQL data layer with Row-Level Security (RLS) policies and server-side RPC functions to enforce cross-user access control.
- Implemented authentication, online presence, typing indicators, read receipts, and unread-message tracking for real-time communication.
- Built a responsive interface with Tailwind CSS, dark/light mode, message filtering, and real-time UI synchronization.

RESEARCH EXPERIENCE

Summer Research Internship – Tripura University Jul 2026
Research Assistantship / 4 Weeks

- Contributed to a PhD-led research initiative on Layer 2 blockchain scalability through systematic analysis of rollups, sidechains, and state channels.
- Evaluated throughput, latency, and gas cost across Layer 2 scaling mechanisms and organized findings into structured comparative benchmarks.
- Synthesized research findings to identify scalability constraints, performance trade-offs, and architectural differences among Layer 2 approaches.

CORE COMPETENCIES

Machine Learning | Deep Learning | NLP | Computer Vision | Predictive Modeling | Data Preprocessing | Feature Engineering | Model Evaluation | Statistical Analysis | Technical Research | Performance Benchmarking